CLASS 10 — MATHS · FORMULA SHEET

Chapter 2: **Polynomials** | Quick revision, one page

Polynomials and degree

- A polynomial $p(x) = a_n x^n + \cdots + a_1 x + a_0$; degree = highest power of x.
- Linear: $ax + b$ (degree 1). Quadratic: $ax^2 + bx + c$, $a \neq 0$ (degree 2). Cubic: degree 3.

Zeros of a polynomial

- A real number k is a zero of $p(x)$ if $p(k) = 0$.
- Zeros are the x-coordinates of the points where the graph of $y = p(x)$ meets the x-axis.

Zeros and graphs

- A linear polynomial has exactly one zero; its graph is a straight line cutting the x-axis once.
- A quadratic polynomial has at most 2 zeros; its graph is a parabola (opening upwards if $a > 0$, downwards if $a < 0$).
- Parabola cuts the x-axis at two points: 2 zeros. Touches it at one point: 1 (repeated) zero. Does not meet it: no real zero.
- A cubic polynomial has at most 3 zeros.

Relationship between zeros and coefficients (quadratic)

- If α, β are the zeros of $ax^2 + bx + c$:
- $\alpha + \beta = -\dfrac{b}{a}, \qquad \alpha\beta = \dfrac{c}{a}.$
- Polynomial from zeros: $k\,[x^2 - (\alpha + \beta)x + \alpha\beta]$, $k \neq 0$.
- Useful: $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$, $\dfrac{1}{\alpha} + \dfrac{1}{\beta} = \dfrac{\alpha + \beta}{\alpha\beta}$.

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